

5(a)	<i>any four from:</i> <i>mark as pairs</i> <i>Name 1</i> dNTPs / <u>DNA</u> nucleotides ; <i>Explanation 1</i> to, anneal / bind, to complementary bases ; <i>Name 2</i> Taq polymerase ; <i>Explanation 2</i> to make, phosphodiester bonds / the sugar phosphate backbone / double-stranded DNA / a complementary strand of DNA or <i>ref. to</i> functioning at higher temperature / does not denature at the higher temperature / is thermostable ; <i>Name 3</i> buffer ; <i>Explanation 3</i> to provide, optimum / suitable / constant, pH for, polymerase / DNA or <i>prevents</i>, denaturation of polymerase / damage to DNA ;	4
5(b)(i)	<i>any two from:</i> 1 (DNA ladder) compared to (other fragments) ; 2 <i>ref. to</i> (estimate) length of (DNA) fragment / to identify the fragment number ; 3 <i>ref. to</i> show electrophoresis worked correctly ;	2

Question	Answer	Marks
5(b)(ii)	any three from: 1 (PCR) amplifies / makes many copies of, DNA (fragments) ; 2 (PCR) <i>ref. to</i> selecting fragments, small enough / less than 1000 bp / not more than 916 bp (for sequencing) ; 3 (electrophoresis) separate the (DNA) fragments ; 4 (electrophoresis) confirms / identifies, fragment lengths / fragment number / PCR worked correctly ;	3
5(c)	$(2^{35} =) 3.4 \times 10^{10}$;	1
5(d)	any four from: 1 child may have disease when, neither parent has the disease / no family history ; 2 early, detection / treatment (for gene therapy to work) ; 3 improve, quality of life / life expectancy ; 4 <i>ref. to</i> informs family planning (for future pregnancies) ; 5 <i>idea that</i> it does not involve additional sampling ; 6 (screening is) expensive ; 7 treatment / gene therapy, expensive ; 8 test may not always be accurate ; 9 long-term effect of treatment, unknown / negative ;	4